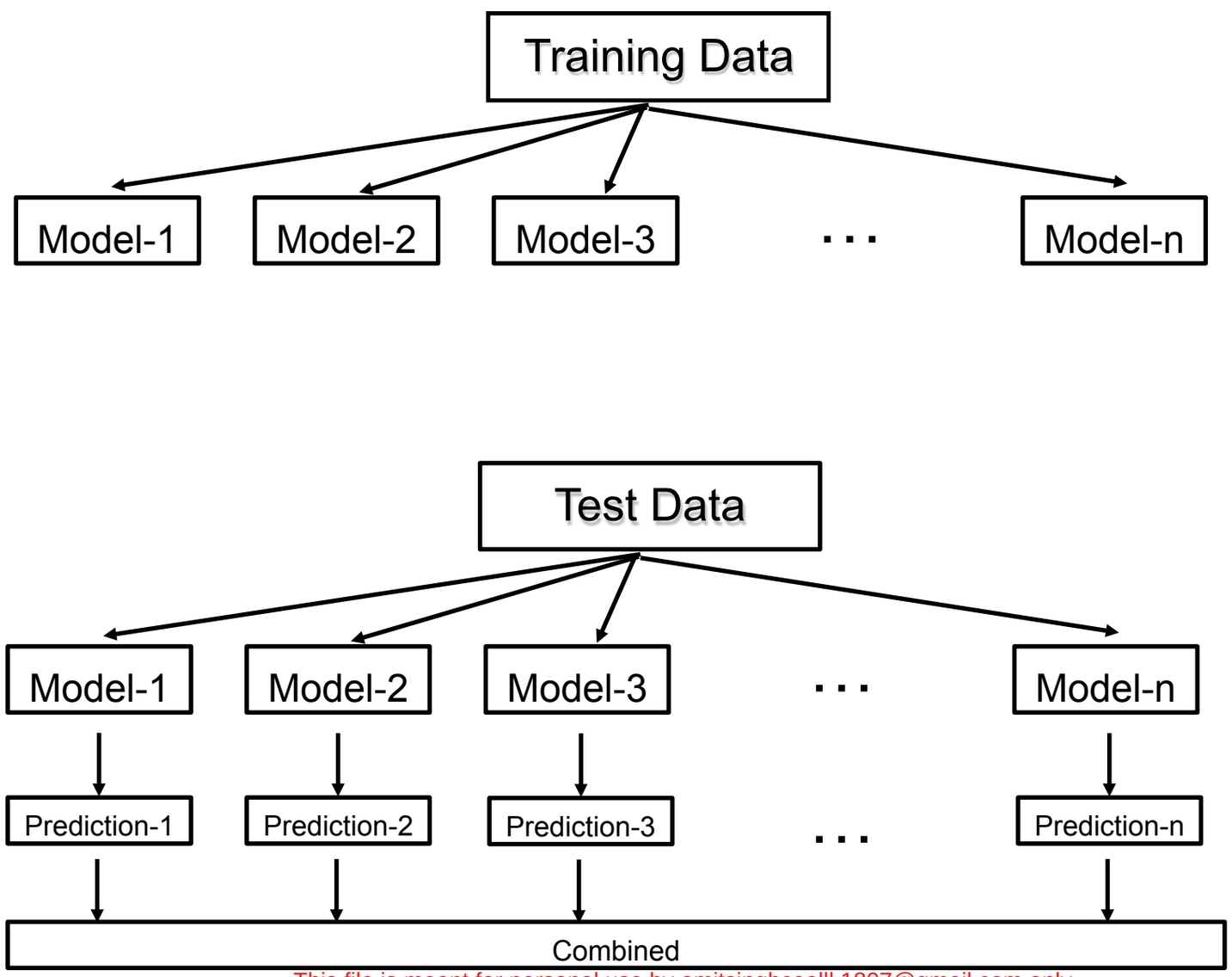
Ensemble Methods

- Ensembles are machine learning methods for combining predictions from multiple separate models.
- The central motivation is rooted under the belief that a committee of experts working together can perform better than a single expert.



Prediction

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Ensemble Methods

Truth	90% M1	90% M2	90% M3	90%		90% M10	
Y	✓	✓	✓	✓	X	✓	✓
Y	X	X	X	X	X	X	✓
N	✓	✓	✓	X	X	X	✓
..	✓	X	✓	X	X	✓	✓
...	✓	✓	✓	✓	.....	.....	✓
...	✓	✓	✓	✓	✓	.....	...
Y	✓	✓	✓	✓		✓	...
N	✓	✓	✓	✓	✓	✓	...

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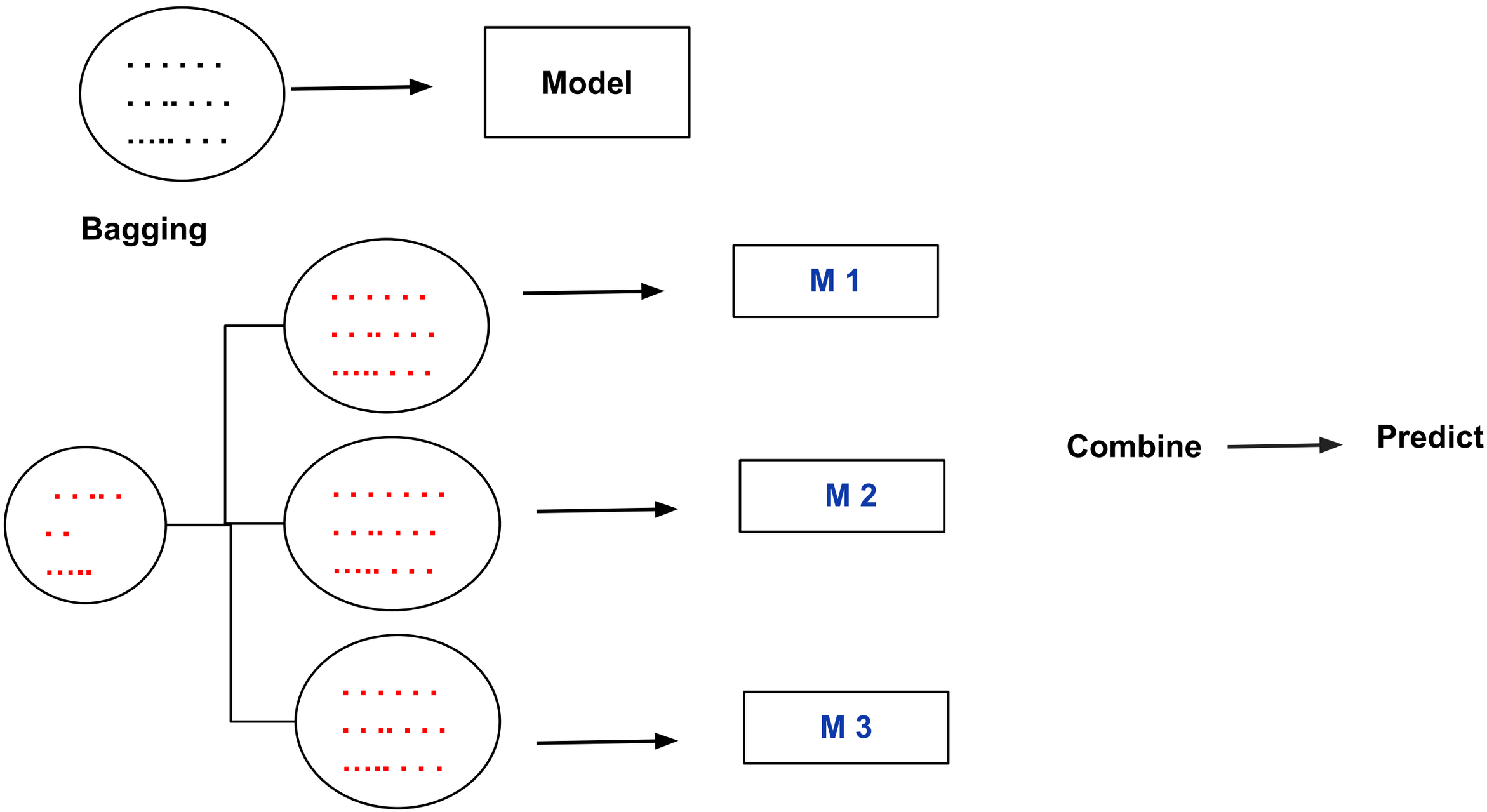
Bagging



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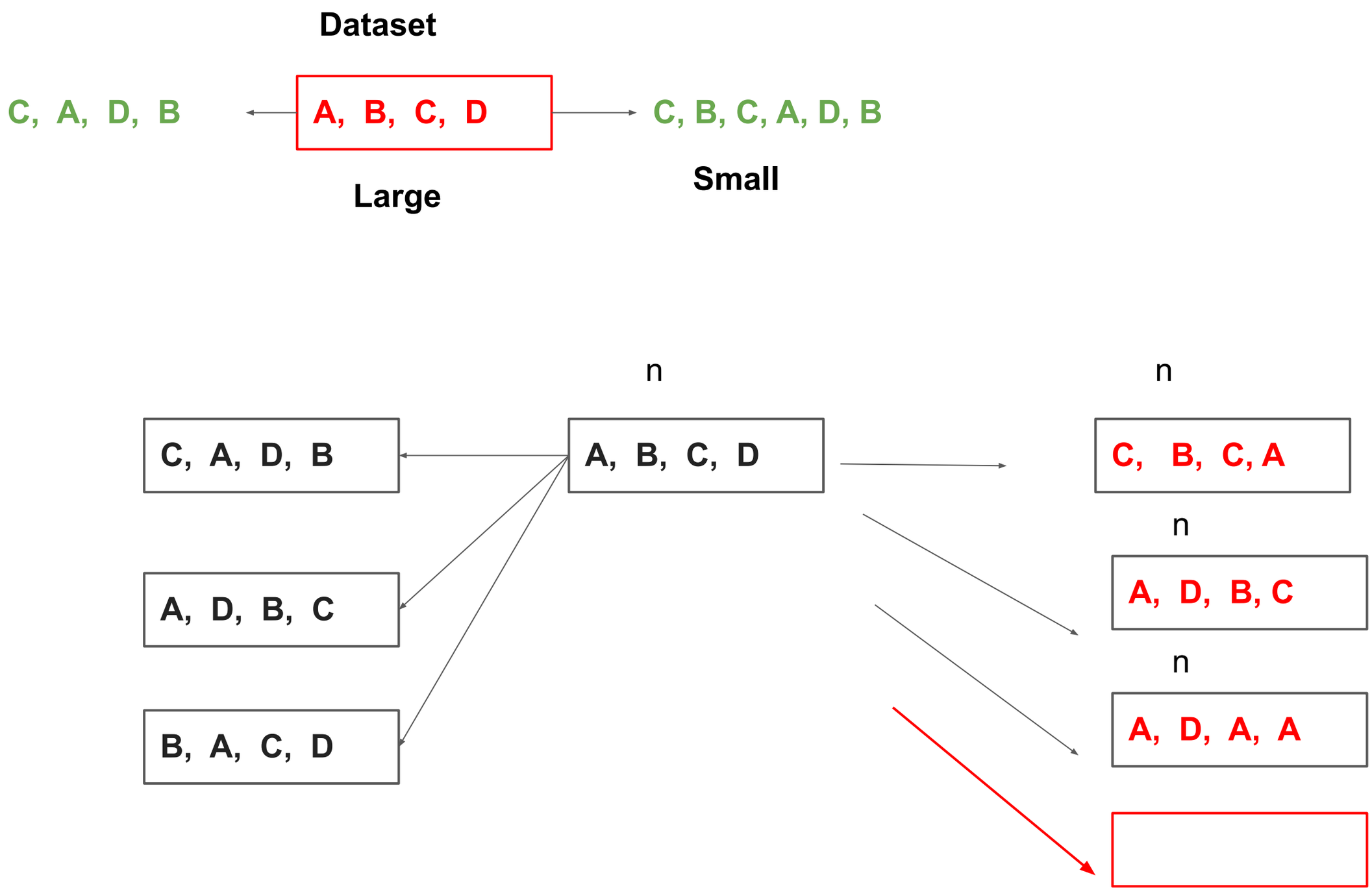
Why Sampling with Replacement?



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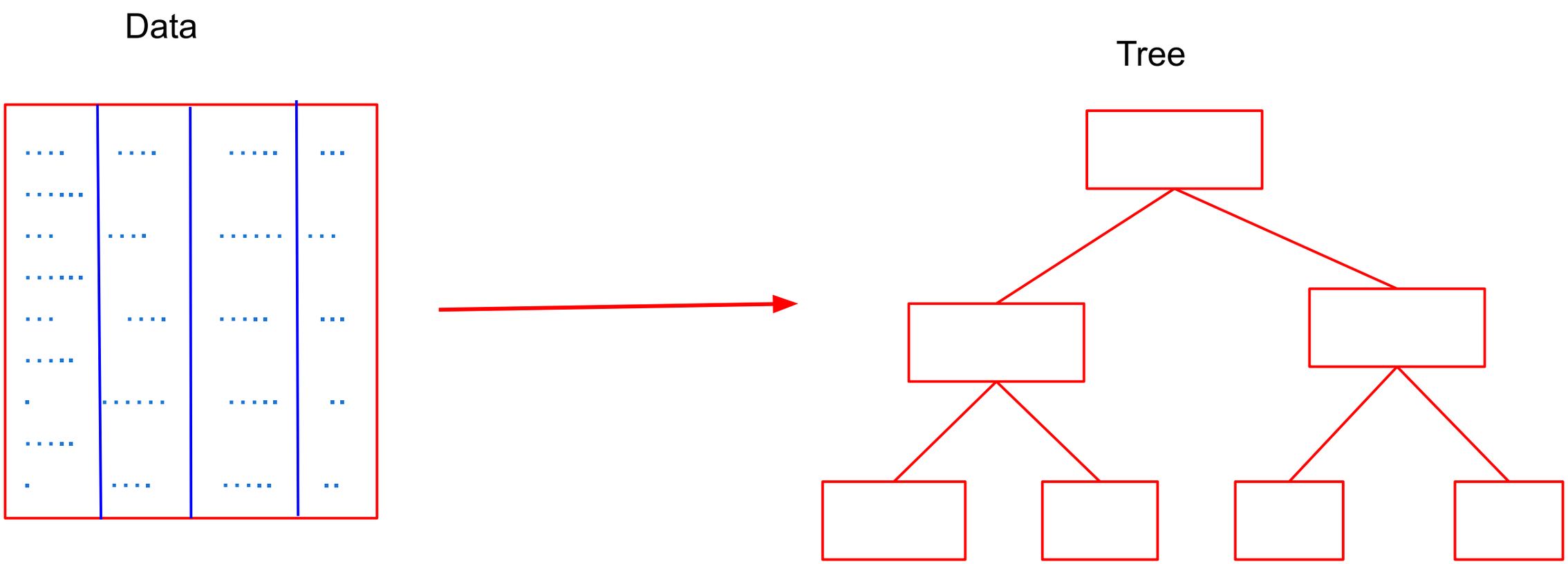
Random Forest



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Tree to a Forest

- Decision trees are very sensitive to even small changes in the data - usually called unstable.
- Can we get a whole bunch of decision trees to work together to yield a better and more robust prediction?
- Then for prediction we could use the mean for regression trees and mode for classification trees
- While individual trees are tend to over-fit training data, averaging corrects this.

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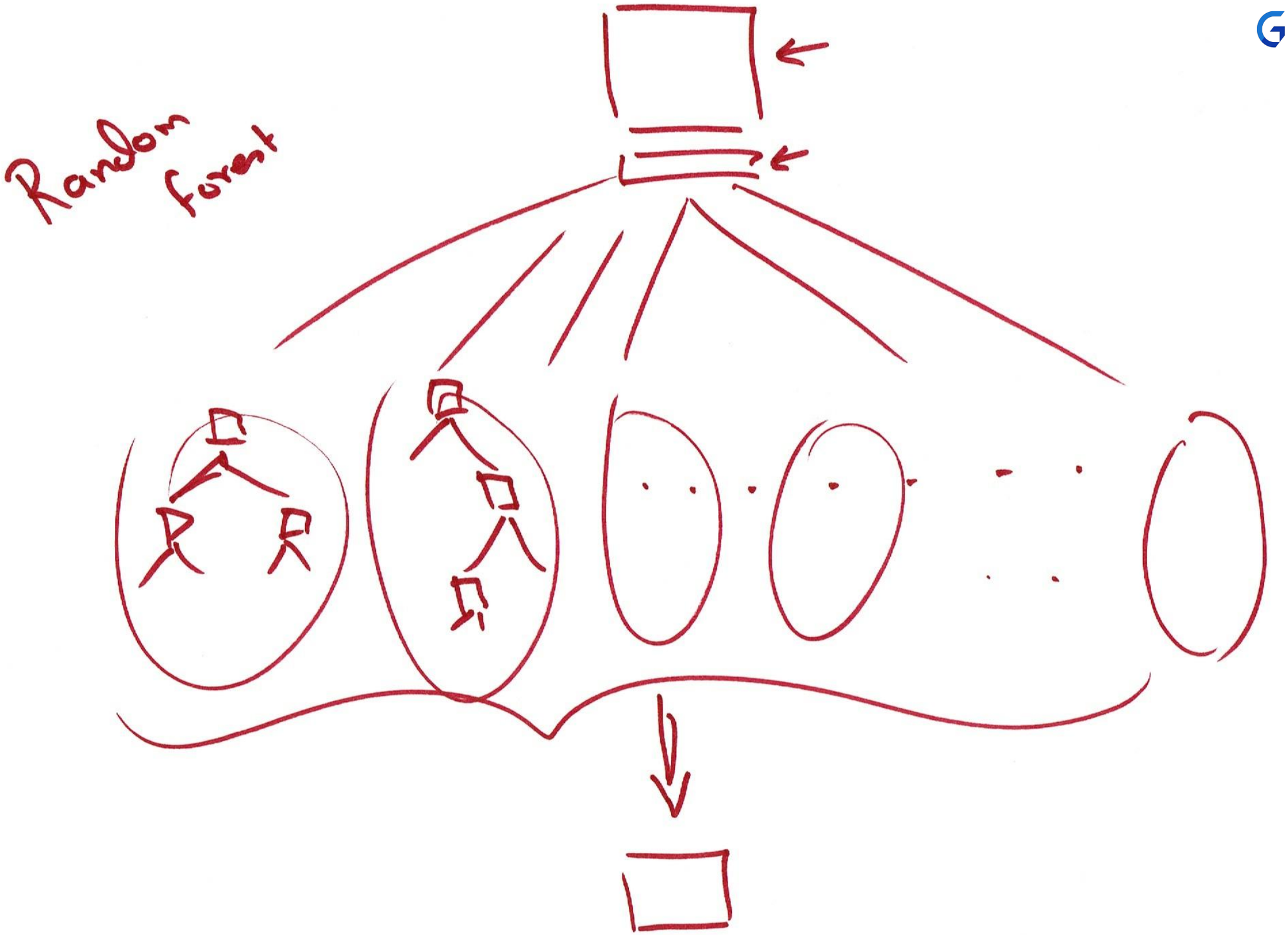

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The General Ideas

- The general procedure of using multiple models (trees, in this case) to obtain better predictive performance is called ensemble learning.

- Bootstrap aggregating. also called bagging:

  - Generate new training subsets of the original, each of the same size (usually the size of the data) by sampling with replacement.

  - By sampling with replacement, some observations may be repeated in each subset.

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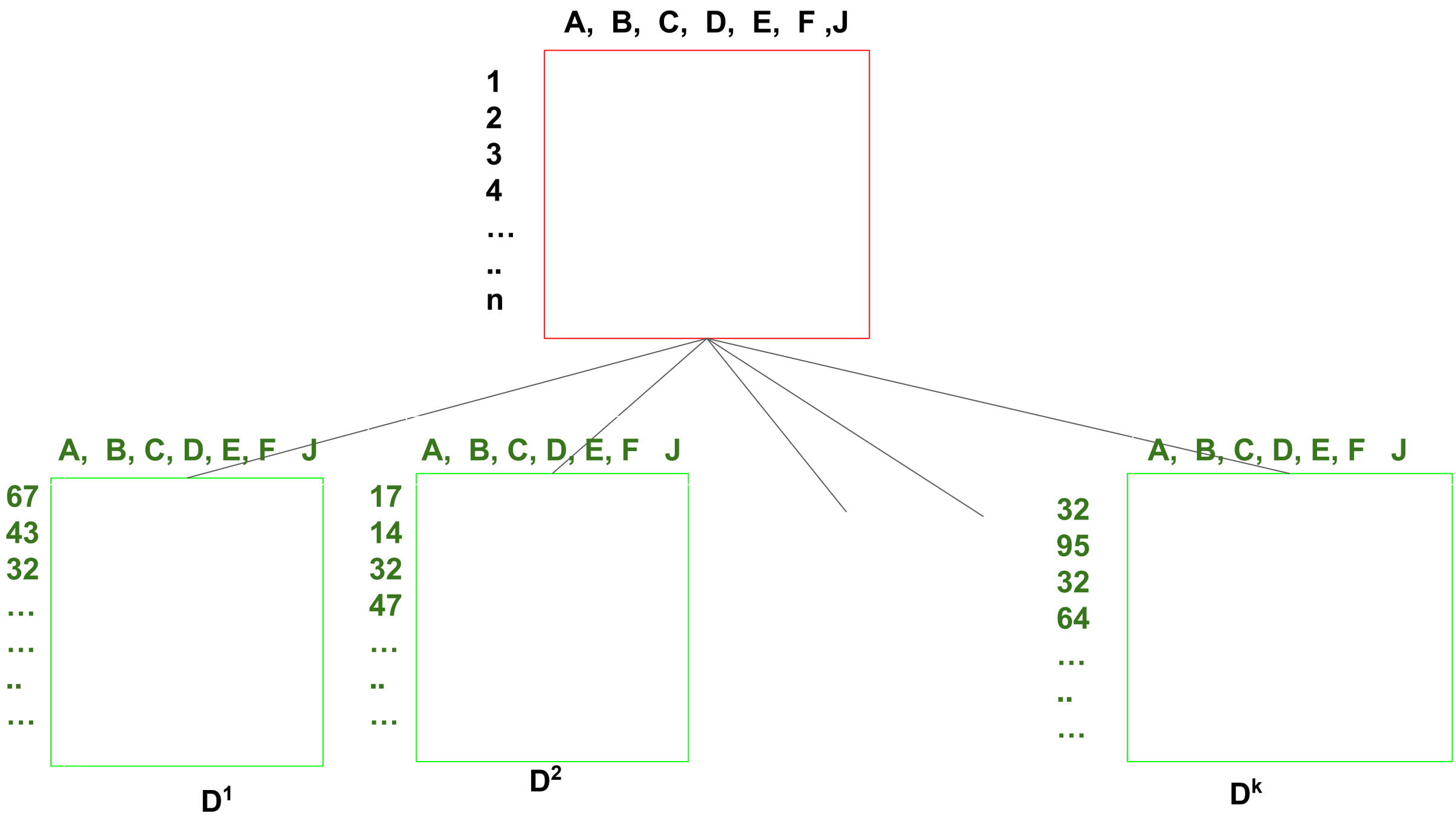
Random Forest



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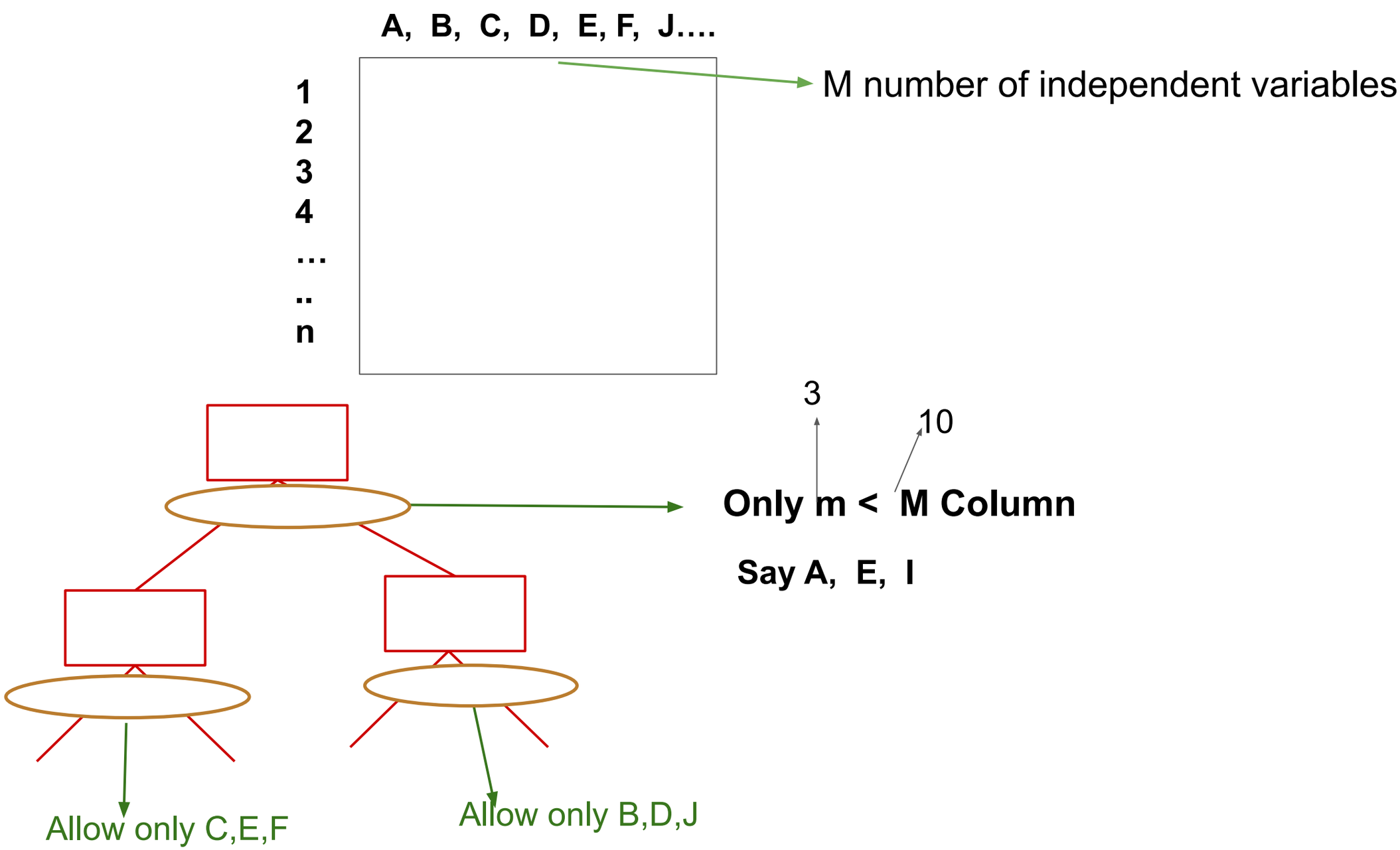
Random Forest



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Random forests

- Random Sampling with replacement
- For each subset build a decision tree. However, only use **m** $< m$ randomly pick independent variables for each node's branching possibilities.
  - Do not prune
- While predicting:
  - Use each tree to make individual predictions
  - Combine predictions using voting:
    - Means for regression
    - Modes for classification

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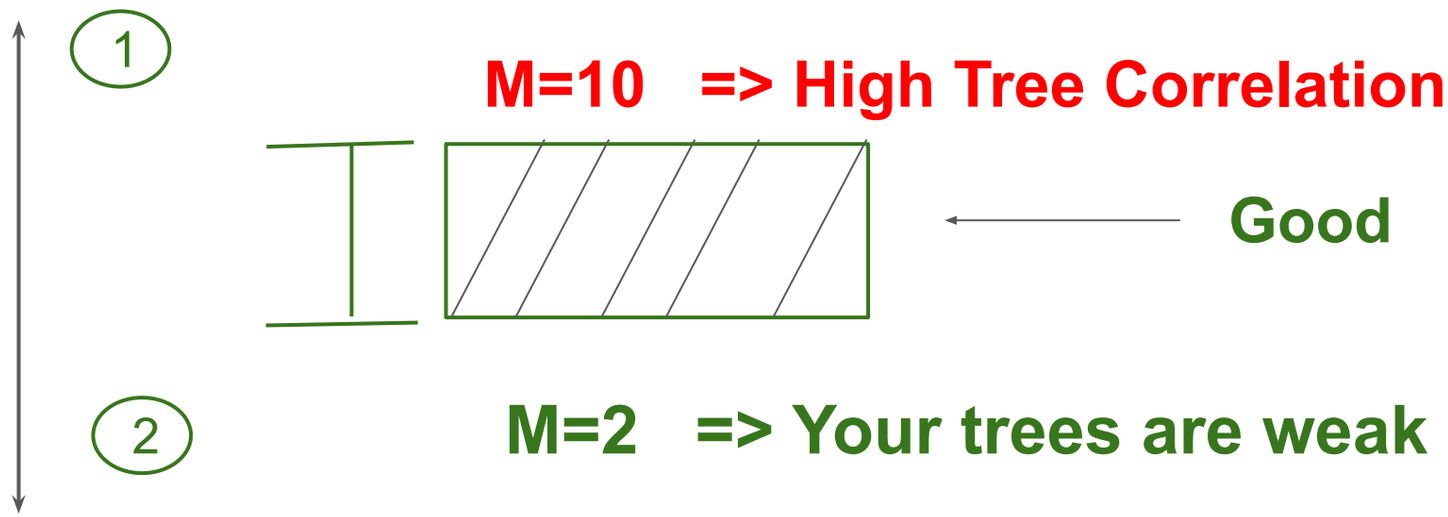
Random Forest

Say M = 10 => A B C D ◯ ◯ ◯ J



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